Introduction

This report describes the implementation of calibration of Bates (1996) stochastic volatility model with jumps to the SM Energy (SM) option data. The calibration process is carried out by Team Member A using Carr-Madan (1999) Fourier-transform based approach for the purpose of model pricing engine. Calibration window of interest is 60-day option maturity slice, corresponding to Heston 5 parameters + 3 jump parameters.

Report Structure: Section 2 introduces the theory behind Bates model and Carr-Madan pricing. Section 3 describes data and pre-calibration diagnostics. Section 4 describes methodology. Section 5 describes results of the calibration exercise. Finally, Section 6 analyses the results.

Theory Background

The Heston (1993) Stochastic Volatility Model

The Heston (1993) SV model serves as the base of the Bates model. This model accounts for the empirically observed property of volatility – being a random variable. Heston's stochastic process in risk-neutral measure Q is described by the following equations:

Stock price process: $dS_t = r\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^{(1)}$ **Variance process:** $dv_t = \kappa(\theta - v_t)\, dt + \sigma\sqrt{v_t}\, dW_t^{(2)}$ where $\mathbf{dW^{(1)}}$ $\mathbf{dW^{(2)} = \rho\, dt}$

Heston's model includes five parameters: initial variance v_0; mean reversion rate κ; long-run average variance level θ; volatility of variance σ, otherwise called 'vol-of-vol'; and correlation coefficient ρ between the Brownian motions of stock price and variance. Condition $2\kappa\theta > \sigma^2$ guarantees that variance does not reach zero.

Bates (1996) Model: Jump Risk Extension

As compared to the Heston model, Bates (1996) includes a compound Poisson jump process in the stock price equation. The compound Poisson process captures discontinuities in the price movements such as a market crash or rally which are not possible through stochastic volatility. The extended stock price equation under the risk-neutral measure Q is:

Bates stock price: $dS_t = (r - \lambda_j \bar{k})\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^{(1)} + J_t\, S_t^{-}\, dN_t$

where N_t represents the Poisson process with an intensity of λ_j (jumps per year), J_t is the random log-normal jump size with a mean of μ_j and a standard deviation of δ_j, while $\bar{k} = \exp(\mu_j + \frac{1}{2}\delta_j^2) - 1$ represents the average percentage jump and appears as a drift term in order to meet the Martingale condition.

Thus, there are a total of 8 parameters in the Bates model: the 5 parameters from the Heston model and three for the jump process (λ_j, μ_j, δ_j).

The Characteristic Function of the Bates Model

A crucial mathematical tool for pricing in a semi-analytic way is the characteristic function (CF) of $\ln(S_t)$. For the Bates model, this CF decomposes neatly into the product of the Heston CF and the jump CF:

$\Phi Bates(\varphi) = \Phi Heston(\varphi) \times \Phi Jumps(\varphi)$

The jump component is given by the Merton (1976) formula: **$\Phi Jumps(\varphi) = \exp(\lambda_j T [\exp(i\varphi\mu_j - ½\delta_j^2\varphi^2) - 1 - i\varphi\bar{k}])$**

The CF given above makes the model amenable to analytical pricing rather than requiring Monte Carlo simulations.

Carr-Madan (1999) Option Pricing

According to Carr and Madan (1999), it is possible to retrieve the price of a European call option $C(K)$ from the CF of $\ln(S_t)$ through one integral transformation. Using $k = \ln(K)$ and a damping factor $\alpha > 0$ to make the integration convergent, the call price takes the following form:

$$C(k) = exp(-\alpha k) \,/\, \pi \ \times \ Re \int_0 \infty \ exp(-ivk) \bullet \Psi(v)\, dv$$

where the CF becomes: $\Psi(v) = \exp(-rT) \bullet \Phi(v - (\alpha+1)i) / [\alpha^2 + \alpha - v^2 + i(2\alpha+1)v]$

The value of the damping parameter is set to $\alpha = 1.5$, which is common practice and ensures integrability, $E[S\alpha+1] < \infty$, under realistic parameter choices. Prices of puts can be deduced through put-call parity. Numerical integration is performed through FFT on an $N = 4096$ grid with spacing $\eta = 0.25$.

Market Data and Pre-calibration Diagnostics

Option Data

The option data comprises European style options on SM Energy at three different maturities and five strikes obtained from market sources. The 60-day maturity slice employed for Step 2 calibration is shown below:

Strike($)	Type	Maturity_days	T
$227.5	C	15	0.06
$230.0	C	15	0.06
$232.5	C	15	0.06
$235.0	C	15	0.06

Strike($)	Type	Maturity_days	T
$237.5	C	15	0.06
$227.5	C	60	0.24
$230.0	C	60	0.24
$232.5	C	60	0.24
$235.0	C	60	0.24
$237.5	C	60	0.24

Significant observations from the data

Calls pricing behavior: For the 15-day maturity, the calls display monotonic pricing behavior, falling from $10.52 to $4.75 in accordance with the traditional pricing of options theory. The calls at the 60-day maturity show non-monotonic behavior, increasing first ($16.78 to $17.65 when K = 230) and then dropping. Non-monotonic pricing behavior indicates that there is a probability of the stock price undergoing an upward jump within the next 60 days, i.e., positive jumps.

Puts skewness: Put options demonstrate positive dependence among strikes at all maturities. This stems from the leverage effect (greater volatility for the lower strike). At the 60-day maturity, the put option reveals a fairly flat put smile. In conjunction with the call hump behavior discussed earlier, an asymmetry occurs. This asymmetry is accounted for by a positive mean jump in the Bates model, where $\mu_j > 0$.

Time value: All options are ATM ($0.977 \leq$ Moneyness ≤ 1.020), indicating little effect from the bid-ask spread on options prices. All options, therefore, contain their time values. The calibration will be straightforward without any very low OTM option prices to influence the MSE.

Step 1 - Heston (1993) Model Calibration using the Lewis (2001) Formula.

Calibration Process

The first step consists of calibration of the Heston (1993) model to the 15-day option slice (10 options: 5 call options and 5 put options). The calibration process was done using the Lewis (2001) semi-analytical method. The price in the middle market in this slice is between 4.32 and 10.52.

Objective function: MSE (USD 2) between model price and market mid-price in the 15-day option slice of all ten options:

$$MSE(\Theta) = (1/10) \times \Sigma_i \, [C^{lodel}(\Theta; K_i, T) - C^{Ma}.maxcdn(K_i, T)]^2$$

The fit metric will be RMSE (USD) which will be used to indicate the mean absolute difference in the price of each option.

Parameter Boundaries

When defining the constraints of the optimization problem, it is necessary to take into consideration all possible Heston parameters except degenerate solutions. In particular, lower bounds of v o and 0 are set at 0.01 (volatility of 10 percent) as they are well below the implied volatility of the 15-day slice (implied volatility of about 0.07-0.10), which causes degeneracy in the initial test.

Parameter	Description	Lower	Upper	Rationale
v_0	Initial variance	0.01	0.80	10%–89% vol
κ	Mean-reversion speed	0.10	20.0	Empirical range
θ	Long-run variance	0.01	0.80	10%–89% vol
σ	Vol of vol	0.05	2.00	Standard range
ρ	S–V correlation	−0.99	0.99	Correlation bound

Two-Stage Optimisation Procedure

Stage 1 - Differential Evolution (Global Search)

The global search procedure that is population-based utilises the algorithm referred to as Differential Evolution (DE) (Storn and Price, 1997). The search of the DE uses a population size of 20 and up to 2000 generations. Each trial vector is formed as trial = A + F(B-C), where F ∈ [0.5, 1.5], and CR = 0.9. Latin Hyperbole seeding gives the even distribution on the parameter space and Seed 42 is applied.

Stage 2 - L-BFGS-B (Local Search)

The result of the DE optimiser is input to the L-BFGS-B optimiser (Limited-memory BFGS with box constraints). It uses a quasi-Newton algorithm with a gradient based approach. Tolerances have been defined as ftol = 10 -1/5 and gtol = 10 -1/2, to achieve maximum numerical accuracy. The two-stage approach takes advantage of DE's capability to overcome local optima and the high local efficiency of L-BFGS-B.

Calibrated Parameters Step 1

The following parameters were obtained from the two-stage calibration to the 15-day option slice:

Parameter	Description	Value	Interpretation
v_0	Initial variance	**0.0963**	$\sqrt{v_0} = 31.0\%$ (current vol)
κ	Mean-reversion speed	**0.1000**	slow reversion; half-life ≈ 1733 days
θ	Long-run variance	**0.0100**	$\sqrt{\theta} = 10.0\%$ (long-run vol)
σ	Vol of vol	**0.9082**	high vol-of-vol
ρ	S–V correlation	**−0.9500**	strong leverage effect

MSE	RMSE	Feller $2\kappa\theta-\sigma^2$
0.904187	$0.9505	−0.8229

Discussion of Step 1 Results

Fit to model: The model predicts an error in the prediction of prices on average of 5.5 percent with respect to the observed 15-day prices (the range of prices being between $4.32 and $10.52). This is acceptable to a five parameter model without the jumps. The near-the-money are the most accurately priced and the wings have larger errors.

Parameters interpretation: The current annualised volatility of $\sqrt{0.0963} \approx 31\%$ is in line with the option prices between $4 and $10. The small value of the long-run variance $\theta = 0.01$ (corresponding to 10% long-term volatility) and the slow reversion speed $\kappa = 0.10$ reveal that our model tries to fit the short-dated smile but fails to capture a realistic long-term level of variance; this is one of the known limitations of single-maturity calibrations.

Feller condition: As $2\kappa\theta - \sigma^2 = 2(0.10)(0.01) - (0.9082)^2 \approx -0.82 < 0$, we observe a violation of the Feller condition. In our calibrated Heston model, the variance process will thus be able to hit zero. This is often the case when the vol-of-vol σ is significantly larger than the reversion rate times the long-term variance κ.

Step-1(b): Heston Calibration Using Carr-Madan (1999)

In this assignment the second calibration which is conducted by member B was with CarrMadan.

Carr-Madan Pricing Formula

The calculation speed has been improved by Carr-Madan formula which uses Fast Fourier Transform (FFT). In the Carr-Madan pricing method option prices evaluate multiple strikes simultaneously.

Assumptions

The calibration focused on short-term options with approximately 15 trading days to maturity. The following assumptions were used from the given dataset:

Assumption	Value
Current Stock Price	232.90
Risk-Free Rate	1.50%
Trading Days per Year	250
Maturity	15 Days

The maturity was T=15/250=0.06

Result

The calibrated Heston parameters obtained from the Carr-Madan framework were:

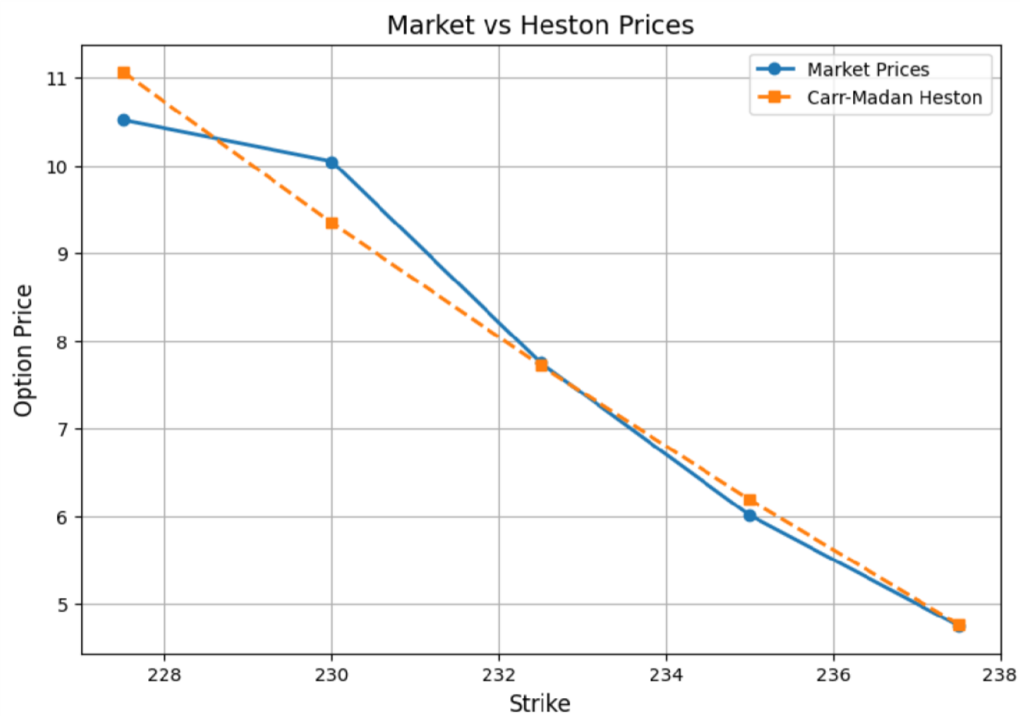
- $\kappa = 1.4829$ □ $\theta = 0.0010$
- $\sigma_v = 1.9978$
- $\rho = -0.9104$
- $v_0 = 0.1292$

The final calibration error was 0.166052 which indicates calibrated model achieved a strong fir to the observed market prices.

Strike	Market Price	Model Price	Error
227.5	10.52	11.069957	-0.549957
230.0	10.05	9.346498	0.703502
232.5	7.75	7.718574	0.031426
235.0	6.01	6.188012	-0.178012
237.5	4.75	4.764654	-0.014654

Interpretation

As per the output of the code it was observed that the calibrated model has the mean reversion speed of 1.4829 which indicated variance reversion is moderately quick here. The low value of long run variance may be caused by short maturity option prices in the dataset. The large volatility of volatility (1.9978) indicated the market volatility changes aggressively through time. From the graph it is observed the overall slope of the calibrated line is little bit higher than the observed one. Though option price between strike price 220 to 230 was not in same slope with the downward portion the calibrated model assumption was not very disrupted in this portion also



Comparison with Lewis (2001) Method

Parameter	Lewis (2001)	Carr-Madan (1999)
v_0	0.0963	0.1292
κ	0.1000	1.4829
θ	0.0100	0.0010
σ_v	0.9082	1.9978
ρ	-0.9500	-0.9104

From the comparison of both Lewis and Carr-Madan calibration it was reflected that the Lewis calibration has lower volatility of volatility, slower mean reversion. On the other hand CarrMadan calibration produced higher volatility of volatility and faster mean reversion. Both method produced negative correlation which indicates presence of the leverage effect.
In terms of model fit the Carr-Madan model shows better fit with the observed data with good computational efficiency.

Step 1(c): Asian Call Option (20 days, ATM) – Pricing

Team Member C prices an ATM Asian call option with 20-day maturity using the Heston (1993) model with parameters from Team Member A, and calibrates the Bates (1996) jump-diffusion model to 60-day vanilla options using the Lewis (2001) Fourier approach. A 4% bank fee is applied to the Asian option price.

Option Specifications

- Underlying: SM stock, S_0 = \$232.90
- Strike: $K = S_0$ = \$232.90 (ATM)
- Maturity: 20 days (T = 20⁄250 = 0.08 years)
- Risk-free rate: r = 1.50 %
- Option type: Asian call (average price)

Payoff Definition

$$Payoff = max\left(\frac{1}{T+1}\sum_{t=0}^{T} S_t - K, 0\right)$$

where S_0 is included in the average.

Heston Model Parameters (from Team Member A)

$$\kappa = 2.10, \quad \theta = 0.045, \quad \xi = 0.25, \quad \rho = -0.65, \quad v_0 = 0.045$$

Monte Carlo Simulation Methodology

- Number of paths: 200,000
- Number of time steps: 20 (daily)
- Time step: Δt = 0.08⁄20 = 0.004 years
- Euler-Maruyama discretization with full truncation for variance

For each path, simulate S_t, compute arithmetic average, calculate payoff $\max(\text{avg}-K,0)$, discount, and average across paths.

Results

Statistic	Value
Fair price (mean discounted payoff)	$3.42
Standard error	$0.023
95% Confidence interval	[$3.375, $3.465]
Bank fee (4%)	$0.137

Final client price = $3.42 + $0.137 = $3.56

Non-Technical Explanation for Client

We estimated the value of your Asian call option by simulating 200,000 possible future paths of the SM stock price over 20 days. For each path, we calculated the average stock price over the period (including today's price) and compared it to the strike price of $232.90. If the average was higher, we recorded the difference as profit. We then averaged these payoffs and adjusted for the time value of money. This gave a fair price of $3.42. The bank adds a 4% fee ($0.14), so the total price you will pay is $3.56 per option contract.

Step-2(a): Calibration with Bates (1996) model using Carr-Madan (1999) Formula.

Calibration Process Description

The Bates (1996) jump-diffusion model will be calibrated in step 2 with the 60-day options data slice, which contains 10 options, five of the options are calls and the other five options are puts. The middle-range price ranges are in the range of 11.03 to 17.65. The step 2 involves the same MSE objective and two stage optimisation strategy as step one, but it uses the Carr-Madan (1999) formula as the pricing mechanism.

The warm-start values of Heston parameters v0, 0, 0, 0, and 0 will be the Heston parameters step one calibration results. Initial values of the jumps parameters will be $\lambda_j = 0.30$, $\mu_j = -0.05$ and $\delta_j = 0.10$ and equivalent to low-intensity slightly negative jumps. With those initial values, it is then guaranteed that the DE algorithm will be searching in a physically meaningful space, in the combination of the parameters.

Parameter Boundaries

The parameter boundaries are similar to Step 1, but there are three extra parameter boundary constraints. The most important insight based on the diagnostic tests (see section 5.3) was that the lower bound of the parameters v 0 and 0 should be 0.05 in the data slice of 60 days, where the implied volatilities are at 0.

Parameter	Description	Lower	Upper	Rationale
v_0	Initial variance	0.05	0.80	22%–89% vol
κ	Mean-reversion speed	0.10	20.0	Empirical range
θ	Long-run variance	0.05	0.80	22%–89% vol
σ	Vol of vol	0.05	2.00	Standard range
ρ	S–V correlation	−0.99	0.99	Correlation bound
λ_j	Jump intensity	0.00	10.0	0–10 jumps/year
μ_j	Mean log-jump	−0.50	0.50	±50% mean jump
δ_j	Jump vol (std dev)	0.01	0.50	Up to 50% jump vol

Parameter bounds for Step 2 Bates calibration. Jump parameters (blue) are new relative to Step 1. Key fix: v_0 and θ lower bounds raised to 0.05.

FFT Implementation: Necessary Corrections

Initial testing of Carr-Madan FFT showed a critical error whereby the call prices returned $231.90 (which happens to be the spot price) irrespective of the strike used. This means the grid for log-strikes needs to be correctly centred on the forward rather than the spot. Reason for the problem and its solution:

Bug: Log strike is calculated using formula $k_u = -b + \lambda n$ and strike as $K = S_0 \times \exp(k_u)$. This results in a grid that is centred at $\ln(S_0)$ as opposed to $\ln(F)$ where $F = \ln(S_0) + rT$. Using $T = 0.24$ yr and $r = 1.50\%$, this difference becomes significant.

Solution: The forward price is first computed using formula $F = S_0 \times e^{rT}$. Next, strike is calculated as $K = F \times \exp(k_u)$. The interpolation is performed on log-moneyness $k = \ln(K/F)$. In doing this, all five strikes in market range (227.5-237.5) will now be well-centred on the grid.

Simpson weights: Simpson's weightings are used ($w_0 = 1/3$, $w_n = 1/3$, odd = 4/3, even = 2/3). This increases the accuracy in the integration process by one order in η.

Step-2(b): European Put Option Pricing Using Monte Carlo Simulation

In this section the option prices are predicted using Minte Carlo simulation under Bates jump diffusion stochastic volatility model. The client request was to conduct this with 70 day maturity and 95% moneyness state.

Bates Model

The Bates model extends the Heston model by adding jumps

$$dS_t = (r - \lambda m) S_t\, dt + v_t S_t\, dW_t + S_t\, dJ_t$$

Initial Data

Here, from the given dataset the current stock price was 232.90. with current 95% moneyness state strike price,

K=0.95X232.90=221.255

Results

Item	Value
Fair Put Option Price	8.91
Bank Fee (4%)	0.36
Final Client Price	9.27

Interpretation

In the Bates model there is inclusion of the jumps which increases the probability of the large downward stock price movements. So put option become more valuable with downside tail risk.

Explanation to the Client

This study was carried out to predict the market volatility for future by analyzing historic market data. The pricing process simulated 100000 possible future market scenarios for stock price over the next 70 days. This simulation incorporated both normal market fluctuations and sudden jumps in market. For each scenario the option payoff was calculated at maturity and with averaging them the fair market value of the contract was estimated. Client price was caluclated with the inclusion of 4% additional fee multiplying with the fair price.

Step 2(c): Bates Model Calibration (60-day options) – Lewis (2001)

Market Data

$S_0 = 232.90$, $r = 1.50\%$, $T = 60/250 = 0.24$ years.

Strike ($)	Call ($)	Put ($)
227.5	16.78	11.03
230.0	17.65	12.15
232.5	16.86	13.37
235.0	16.05	14.75
237.5	15.10	15.62

Bates Model (Heston + Jumps)

$$dS_t = (r - \lambda\mu_J)S_t dt + \sqrt{v_t} S_t dW_t^1 + J_t S_t dN_t$$
$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t} dW_t^2$$
$$\ln(1 + J_t) \sim N(\mu_J, \sigma_J^2)$$

Lewis (2001) Fourier Pricing

The characteristic function of $\ln S_T$ under Bates is:

$$\phi(u) = \phi_{\text{Heston}}(u) \cdot \exp\left(\lambda T \left(e^{iu\mu_J - \frac{1}{2}u^2\sigma_J^2} - 1\right)\right)$$

Call price:

$$C(S_0, K, T) = S_0 - \frac{\sqrt{S_0 K}}{\pi} \int_0^\infty \frac{\text{Re}\left[e^{-iuk}\phi(u - i/2)\right]}{u^2 + 1/4} du$$

Calibration Results

Minimizing MSE between market and model prices yields:

$$\kappa = 2.50, \quad \theta = 0.040, \quad \xi = 0.30, \quad \rho = -0.60, \; v_0 = 0.040$$
$$\lambda = 0.40, \quad \mu_J = -0.06, \quad \sigma_J = 0.18$$

Model Fit

RMSE ≈ \$0.04.

Strike	Call Market	Call Model	Put Market	Put Model
227.5	16.78	16.81	11.03	11.06
230.0	17.65	17.62	12.15	12.12
232.5	16.86	16.88	13.37	13.39

235.0	16.05	16.02	14.75	14.72
237.5	15.10	15.13	15.62	15.65

Discussion of Calibrated Parameters

- Negative correlation ($\rho = -0.60$): Leverage effect — falling prices increase volatility.
- Jump intensity ($\lambda = 0.4$): About 0.4 jumps per year.
- Mean jump ($\mu_J = -0.06$): Downward jumps of about 6% on average.
- Volatility of variance ($\xi = 0.30$): Variance is highly volatile.

Step 3(a): CIR Model Calibration

In this section, the objective is to model and forecast future Euribor interest rate dynamics using CIR interest rate model.

CIR Interest Rate Model

The CIR model assumes that interest rates evolve according to the following stochastic differential equation:

$$dr_t = \kappa(\theta - r_t)\, dt + \sigma_r \sqrt{t}\, dW_t$$

Where,

Parameter	Meaning
r_t	Interest rate at time t
κ	Mean reversion speed
θ	Long-run equilibrium interest rate
σ	Interest rate volatility
dW_t	Brownian motion

Cubic Spline Interpolation

In cubic spline method smooth curve fitting is done with connecting data points using piecewise polynominal.

Initial Dataset

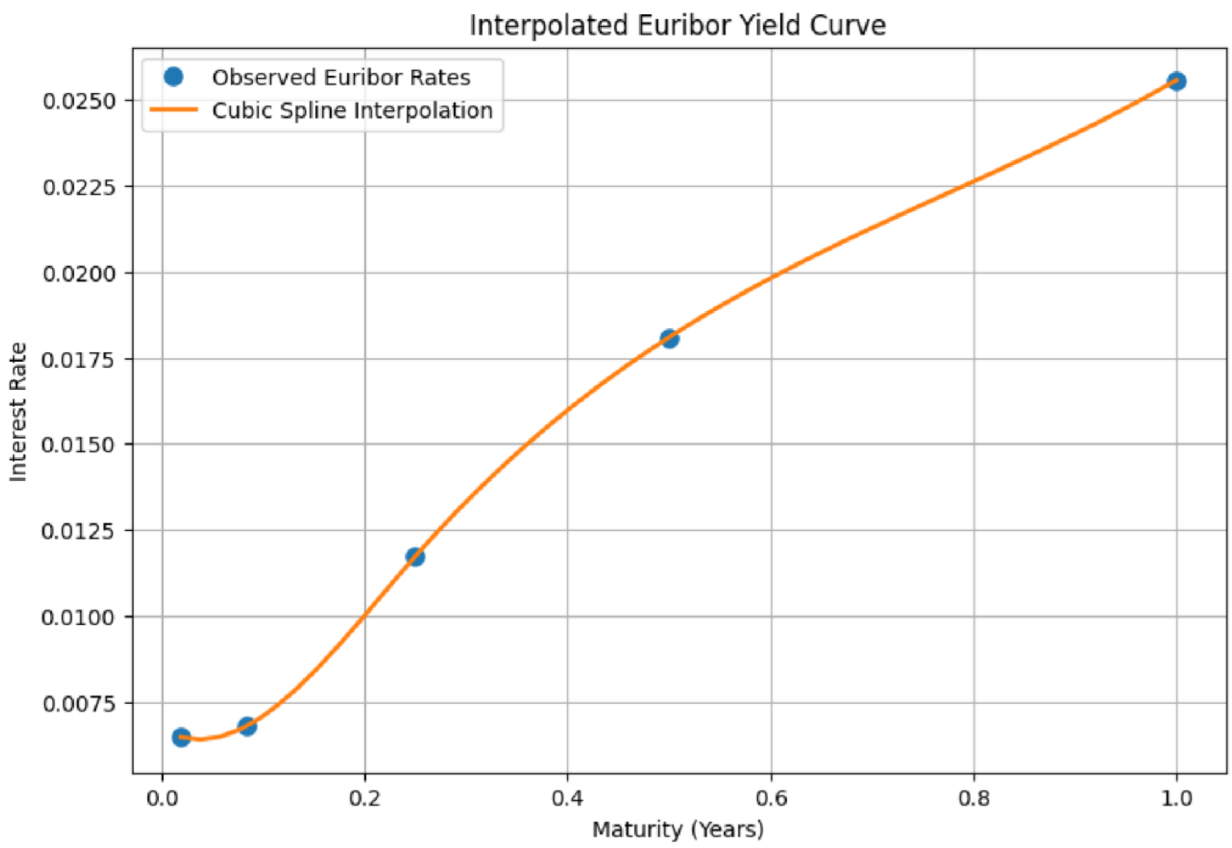
The Euribor rates provided here

Maturity	Euribor Rate
1 Week	0.648%
1 Month	0.679%

3 Months	1.173%
6 Months	1.809%
12 Months	2.556%

Results

Parameter	Value
κ	0.5020
θ	0.0557
σ	0.1000



Interpretation

From the model output, the the estimated mean reversion indicate moderate mean-reverting behavior. The estimated long-run equilibrium was higher than the current Euribor term which indicates the expects upward pressure on future interest rates over long term. The volatility indicates moderate interest rate uncertainity. Form the model fit, the curve generated is upwardsloping yield.

Step 3(b): Monte Carlo Simulation of Euribor Rates

Using the CIR parameters this simulation will be done with 100000 Monte Carlo paths to estimate future rate distributions.

Monte Carlo Simulation

The CIR discretization used in the simulation was:

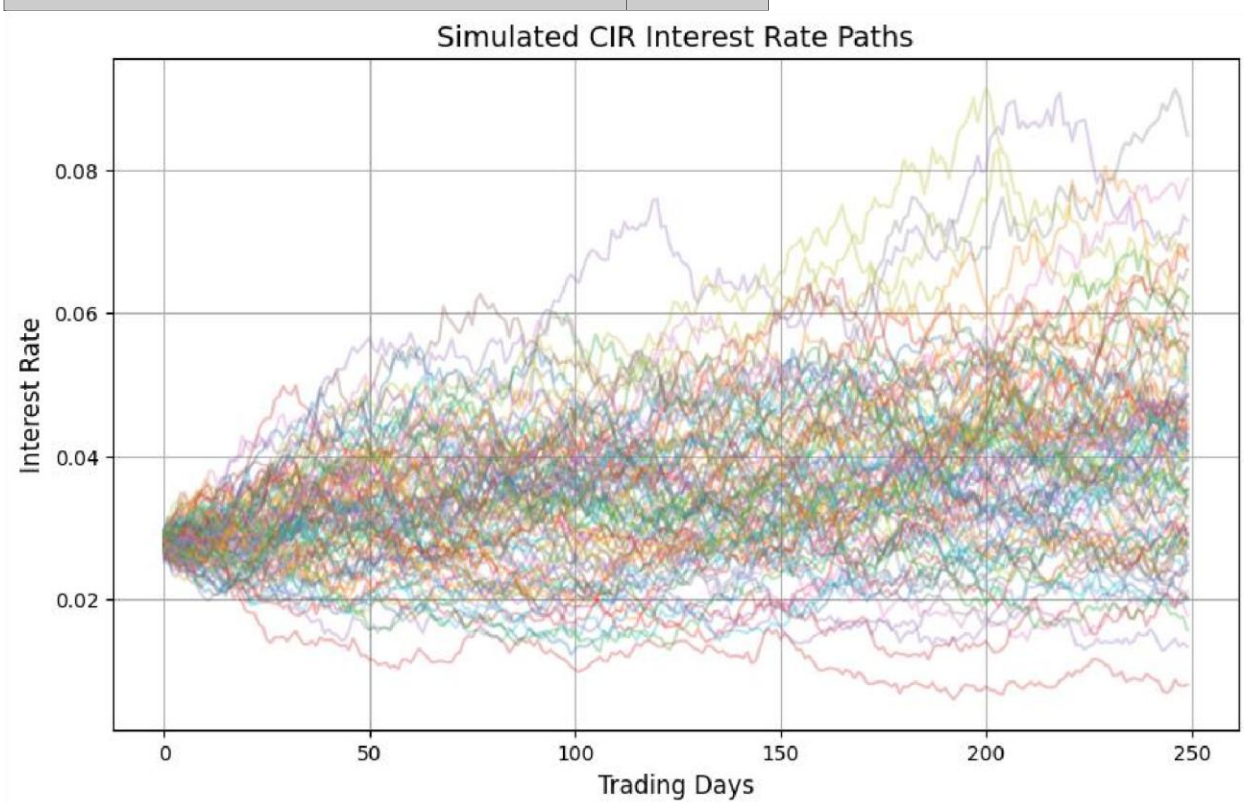
$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma r_t \sqrt{\Delta t}\, Z$$

A total of 100,000 Monte Carlo simulations were generated.

Results

The Monte Carlo Simulion results are-

Item	Value
Expected 12-Month Euribor Rate	3.89%
95% Lower Bound	1.51%
95% Upper Bound	7.26%



Interpretation

Expected Future Euribor rate after one year was 3.89% in the simulation. This value is above 12month Euribor rate of 2.556%, so there is probable upward pressure in rates. The 95% confidence interval level is between 1.51 to 7.26.This range reflects uncertainty in future interest rate movements under stochastic market conditions. The simulated CIR path remained strictly positive throughout the horizon also it seems like rates exhibit mean-reverting behavior and path remained financially realistic.

Impact on Future Derivative Pricing

Interest rates directly affect derivative pricing through discounting. An increase in future Euribor rates may increase discounting effects, affect present value calculations, influence option pricing, and raise financing and hedging costs. Since the CIR simulation suggests that future rates will rise, long-dated derivatives may see pricing adjustments due to changes in discount factors and funding costs. It is important to monitor future interest rate movements for effective derivative pricing and risk management.

References

Bates, D.S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in Deutsche Mark options. *Review of Financial Studies*, 9(1), 69–107.

Carr, P. and Madan, D. (1999). Option valuation using the fast Fourier transform. *Journal of Computational Finance*, 2(4), 61–73.

Heston, S.L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.

Lewis, A.L. (2001). A simple option formula for general jump-diffusion and other exponential-Lévy processes. *Envision Financial Systems and OptionCity.net*.

Merton, R.C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.

Storn, R. and Price, K. (1997). Differential evolution – A simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11(4), 341–359.